

Assignment no. 3 (v.1.0¹): SECURITY POLICY DEFINITION AND IMPLEMENTATION

In this assignment, you must properly configure the ACME network in your group's virtual environment by defining and enforcing a security policy.

While performing the tasks, you must describe the process you followed and the tests you performed to verify your solution.

Hand-in dates

Remember: you must submit the assignment before taking the written exam. The evaluation will consider completeness, clearness, and proficiency.

SECURITY POLICY DEFINITION AND IMPLEMENTATION

"A firewall policy dictates how firewalls should handle network traffic for specific IP addresses and address ranges, protocols, applications, and content types (e.g., active content) based on the organization's information security policies" [1].

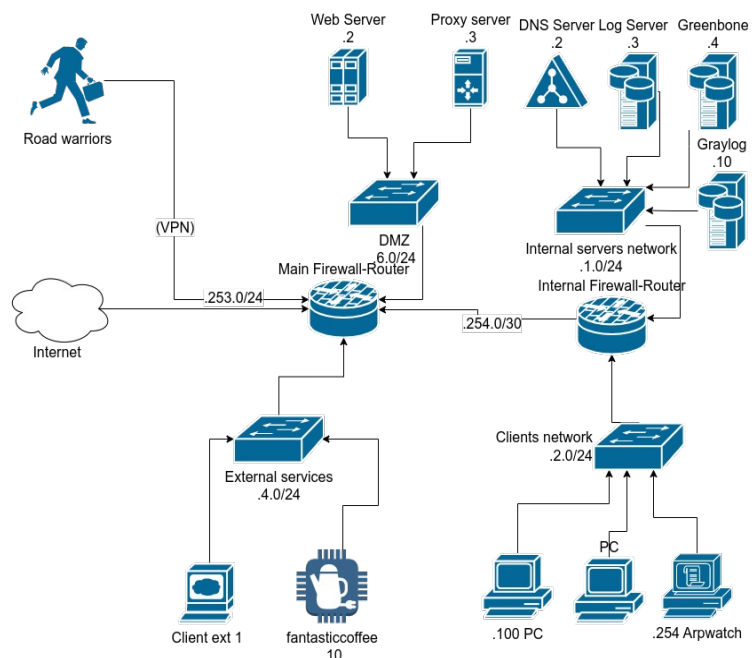
Considering the services you have in your ACME co. network, your group must specify a well-reasoned security policy to protect the overall network properly (see section 4 of [1]).

Once defined, your group has to properly implement the conceived security policy by configuring the firewall rules of the Main Firewall-Router and the Internal Firewall-Router (see section 5 of [1]), and by implementing any other protection on any relevant host.

Once done, the configurations of the two firewalls should be exported (→backup) and included in the submission package (see later) with the names ACME_XX_main.xml and ACME_XX_internal.xml.

Services of the ACME co.

1. A web service on the standard ports (in the Web Server host);
2. A DNS on the standard port (in the DNS Server host);
3. A syslog server in the UDP standard port (in the Log Server host)
4. A proxy server (in the Proxy server host). It will be used by the hosts of the ACME network to request connections via HTTP or HTTPS to the Internet and by the Internet hosts;
5. A log collector (in the Graylog host);
6. A vulnerability scanner (in the Greenbone host).
7. A vending machine in the external services network (in the fantasticcoffee host).



You may need to adequately configure any service mentioned above, even with a simple setting (e.g., HTTPS page or HTTP redirect to HTTPS for the web service).

1) Last change: 20/05/26

Scheme of your hand-in

You have to prepare a document that reports the activities you have performed to realize the assignment tasks. The document should be named `ACME_XX_a3_report.pdf` and should be included together with the exported firewall configurations in **one single** .zip file named `ACME_XX_a3.zip`, without any subdirectory, if not needed (i.e., testing scripts). The zip file is the **only thing** your group has to submit in **classroom**, only one of the members.

The document report should be clear and concise (few pages, please) and have at least the following pieces of information:

1. Group number
2. Student names and numbers
3. Initial brainstorming (where you write your considerations about what to do and how)
4. Configurations of the services
5. Definition of the security policy
6. Policy implementation
7. Test of the policy enforcement
8. Final remarks

References

[1] Scarfone, K. and Hoffman, P. (2009), *Guidelines on Firewalls and Firewall Policy*, Special Publication (NIST SP), National Institute of Standards and Technology, Gaithersburg, MD, [online], https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=901083 (Accessed April 17, 2026)